

**DEPARTMENT OF CHEMICAL & BIOMOLECULAR ENGINEERING
NORTH CAROLINA STATE UNIVERSITY**

CHE 331 QUIZ #1

This is a 30 minute closed-book quiz. Write your name and answer the quiz problems related to your experiment. Turn in this question sheet with your answer.

Name: _____

Batch Distillation Experiment

Problem 1 (15%) Typically the difference between the LHS and RHS of the Rigorous Rayleigh equation is ~5%. Name two sources of error in the lab that may contribute to this difference.

Problem 2 (20%) What are the benefits and drawbacks of using a high reflux ratio? Name one benefit and one drawback.

Problem 3 (50%) On a ***SINGLE***, clearly labeled figure, sketch the profiles of the following two variables as a function of time for the batch distillation of a methanol-isopropanol mixture at two different reflux settings: moles of liquid mixture in the still and methanol mole fraction of the distillate (there should be a total of four graphs on a ***single*** figure). Assume the same initial charge to the still in the two reflux runs. Be sure to indicate which of the two reflux settings is higher.

Problem 4 (15%) What advantage did using a temperature-controlled refractometer provide in the experiment (compared to a refractometer without temperature control)? Explain.